**Training Vectors**

$x_1$ $x_2$ $x_3$ $x_4$

$y_1$=1 $y_2$=1 $y_3$=-1 $y_4$=-1

**Test Vectors**

$T_1$ $T_2$ $T_3$

Use C=1

$$\alpha_1 y_1 + \alpha_2 y_2 + \alpha_3 y_3 + \alpha_4 y_4 = 0$$

$$0 \le \alpha_1 \le C, \ 0 \le \alpha_2 \le C, \ 0 \le \alpha_3 \le C, \ 0 \le \alpha_4 \le C$$

Minimize $\mathcal{L}_D(\alpha) = \frac{1}{2}\sum_{i=1}\sum_{j=1}\alpha_i\alpha_j y_i y_j \left[x_i^T x_j + 1\right]^2 - \sum_{i=1}\alpha_i$

$x_1$ (1 1 1)

$x_2$ (1 2 2)

$x_3$ (1 2 1)

$x_4$ (1 4 5)

| | $x_1$ | $x_2$ | $x_3$ | $x_4$ |
|---|---|---|---|---|
| $x_1$ | 9 | 16 | 25 | 36 |
| $x_2$ | 16 | 36 | 121 | 144 |
| $x_3$ | 25 | 121 | 289 | 441 |
| $x_4$ | 36 | 144 | 441 | 625 |

| $\lfloor x_i \ x_j + 1 \rfloor$ | | | | |
|---|---|---|---|---|
| 10 | 17 | 26 | 37 | |
| 17 | 50 | 101 | 145 | |
| 26 | 101 | 290 | 442 | |
| 37 | 145 | 442 | 626 | |

Minimize : $\lfloor +4.5\alpha_1^2 + 12.5\alpha_1\alpha_2 - 72\alpha_1\alpha_3 - 84.5\alpha_1\alpha_4 + 12.5\alpha_2\alpha_1 + 40.5\alpha_2^2 - 264.$

**Subject to:**

$-[\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4]$

$\sum \alpha_i y_i = 0$

$\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4 = 0$

$\alpha_i \ge 0$

$\alpha_1 \ge 1, \ \alpha_2 \ge 1, \ \alpha_3 \ge 1, \ \alpha_4 \ge 1$

$\alpha_i \le 0$

$\alpha_1 \le 0, \ \alpha_2 \le 0, \ \alpha_3 \le 0, \ \alpha_4 \le 0$

*Quadratic Equation in α with constraints*